Zero Phase Filtering

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Key Concepts

1) A zero-phase filter introduces no phase distortion, not even the time delay of a linear phase filter.

2) Zero-phase filters are always non causal.

    a) Zero phase filtering can only be accomplished as post processing of stored data.

    b) Zero phase filtering cannot be implemented when processing data in real time (as it is collected).

3) Zero phase filtering is usually performed by filtering the data once, then reversing time by reversing the order of data samples, then filtering again followed by reversing the order of data samples a second time.

    a) Reversing order of the data samples requires all data samples be available.

    b) The magnitude response implemented by this procedure is the square of the magnitude response of the filter because the data is passed through the filter twice.

4) Zero phase filtering blurs the data in both directions, thus allowing the effect of a transient in the data to be seen before the transient occurs.

Video Screenshots

Zero phase: no phase distortion, not even time delay of linear phase

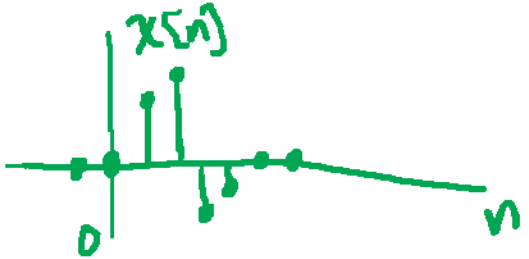
Noncausal filter – post-processing of stored data

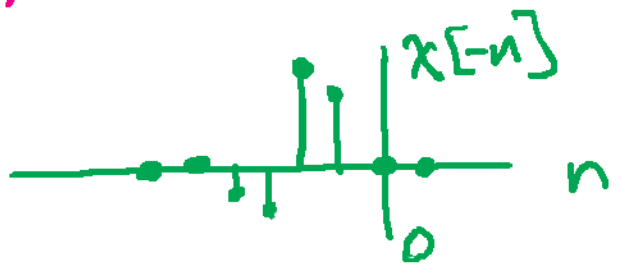
blurs transients forwards and backwards in time

MATLAB command "filtfilt"

Uses time-reversal property of DTFT

If $x[n] \overset{DTFT}{\longleftrightarrow} X(e^{j\omega})$, then $x[-n] \overset{DTFT}{\longleftrightarrow} X^*(e^{j\omega})$





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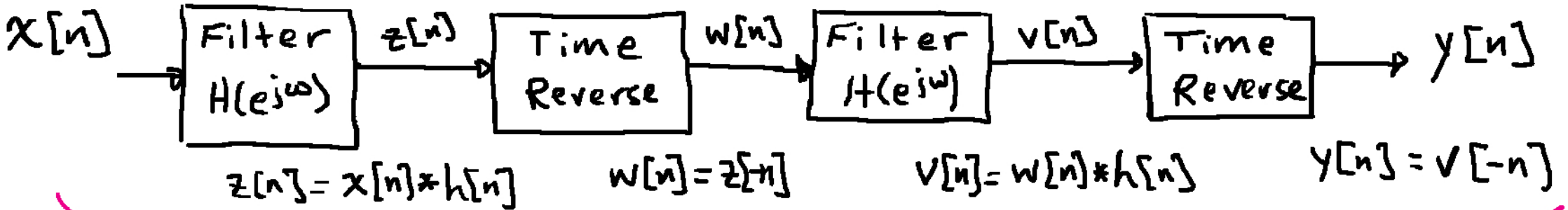
MATLAB command "filtfilt"

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If $x[n] \overset{\text{DTFT}}{\longleftrightarrow} X(e^{j\omega})$, then $x[-n] \overset{\text{DTFT}}{\longleftrightarrow} X^*(e^{j\omega})$

Filter data forwards, then filter backwards in time

Block diagram



1) $Z(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$

2) $W(e^{j\omega}) = Z^*(e^{j\omega}) = H^*(e^{j\omega})X^*(e^{j\omega})$

3) $V(e^{j\omega}) = H(e^{j\omega})W(e^{j\omega}) = |H(e^{j\omega})|^2 X^*(e^{j\omega})$

4) $Y(e^{j\omega}) = V^*(e^{j\omega}) = |H(e^{j\omega})|^2 X(e^{j\omega})$

$$H_{eff}(e^{j\omega}) = |H(e^{j\omega})|^2$$

Issues –

1) Entire input data sequence $x[n]$ must be available

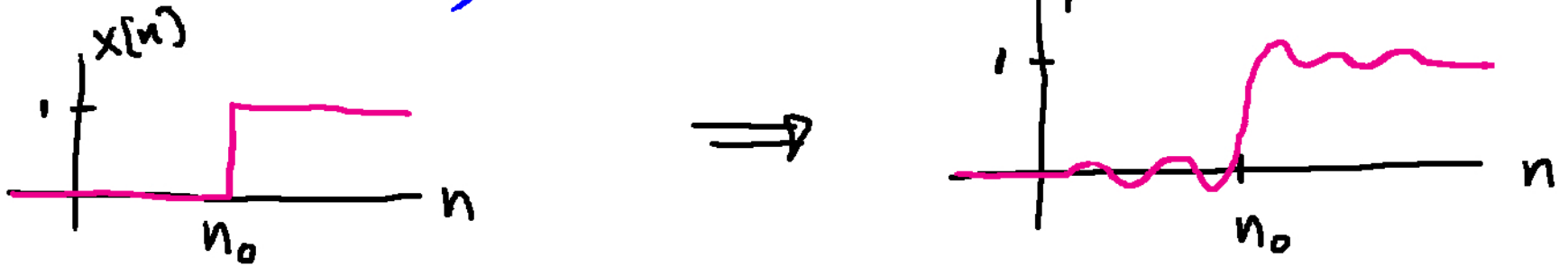
2) Translate specifications on $H_{eff}(e^{j\omega})$ to $H(e^{j\omega})$

$$H_{eff}(e^{j\omega}) = |H(e^{j\omega})|^2$$

Example: 60 dB stopband attenuation for $H_{eff}(e^{j\omega})$

$\Rightarrow$ design $H(e^{j\omega})$ to have 30 dB attenuation

3) Causality is lost



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